

Individual Olympiad Category A

Problem 1

Suppose we are given 2023 displays and 2023 remotes, but we don't know which display corresponds to which remote. In one operation, we can specify an arbitrary set of remotes, and all of them will be switched from **power off** to **power on** or vice versa simultaneously. We will then see which displays come on (initially they are all off). Find the least number of operations to determine which remote corresponds to which display?

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Solution

In the first operation specify a set of $\frac{n}{2}$ remotes. Now we have two sets of $\frac{n}{2}$ remotes, and we know which $\frac{n}{2}$ displays they both correspond to. Now we want to apply the algorithm for $\frac{n}{2}$ displays and remotes to the two sets. Hence it initially appears that we have the recursion $f(n) = 2f\left(\frac{n}{2}\right) + 1$, where $f(n)$ is the number of steps taken by our algorithm for n displays. However, note that we can actually apply the algorithms for both sets simultaneously, since we know which set of remotes corresponds to which set of displays. Thus, the actual recursion is $f(n) = f\left(\frac{n}{2}\right) + 1$. Since $f(1) = 0$, we inductively get $f(n) = \lceil \log_2(n) \rceil$.

Problem 1(Cat A,B) Marking Scheme

No	Steps	Points
1.	Dividing algorithm and observing first recurrence.....	5pts
2.	Observing last recurrence and calculating all parts simultaneously	3pts
3.	Deriving answer $\log(n)$ (Cat A) or showing that 4 steps enough(Cat B).....	2pts

Problem 2

Let $\{a_n\}_{n=0}^{\infty}$ numerical sequence defined by $0 < a_0 < \pi$ and

$$a_n = \frac{1}{n} \sum_{k=0}^{n-1} \arctg(a_k) \text{ for } n \geq 1$$

Prove that the numerical sequence $\{a_n \sqrt{\ln(n)}\}_{n=1}^{\infty}$ is convergent and find its limit.

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Solution

By the recurrence relation

$$a_{n+1} = \frac{\sum_{k=0}^{n-1} \arctg(a_k) + \arctg(a_n)}{n+1} = \frac{na_n + \arctg(a_n)}{n+1}$$

which implies that for $0 < a_0 < \pi$, and note that $\arctg(x) \leq x$

$$0 < a_{n+1} < \frac{na_n + a_n}{n+1} = a_n$$

that is the sequence $(a_n)_n$ is positive and decreasing and therefore it has a limit l which satisfies $(n+1)l = nl + \sin(l)$. Thus, it follows that $l = 0$. Moreover, by Taylor approximation,

$$a_{n+1} = \frac{na_n + a_n - \frac{a_n^3}{3} + o(a_n^3)}{n+1} = a_n \left(1 - \frac{a_n^2}{3(n+1)} + \frac{o(a_n^2)}{n+1} \right)$$

and

$$\frac{1}{a_{n+1}^2} = \frac{1}{a_n^2} \left(1 - \frac{a_n^2}{3(n+1)} + \frac{o(a_n^2)}{n+1} \right)^{-2} = \frac{1}{a_n^2} \left(1 + \frac{2a_n^2}{3(n+1)} + \frac{o(a_n^2)}{n+1} \right) = \frac{1}{a_n^2} + \frac{2}{3(n+1)} + \frac{o(1)}{n+1}$$

Finally, by the Stolz-Cesaro Theorem,

$$\lim_{n \rightarrow \infty} (a_n \sqrt{\ln(n)})^2 = \lim_{n \rightarrow \infty} \frac{\ln(n)}{\frac{1}{a_n^2}} \stackrel{s.c}{\Rightarrow} \lim_{n \rightarrow \infty} \frac{\ln(n+1) - \ln(n)}{\frac{1}{a_{n+1}^2} - \frac{1}{a_n^2}} = \lim_{n \rightarrow \infty} \frac{\ln(1 + \frac{1}{n})}{\frac{2}{3(n+1)} + \frac{o(1)}{n+1}} = \frac{3}{2}$$

Since, $a_n \sqrt{\ln(n)} \geq 0$, we may conclude that $\lim_{n \rightarrow \infty} a_n \sqrt{\ln(n)} = \sqrt{\frac{3}{2}}$.

Problem 2(Cat A) Marking Scheme

Nº	Steps	Points
1.	Manipulations with recurrence $a_{n+1} = \frac{\sum_{k=0}^{n-1} \arctg(a_k) + \arctg(a_n)}{n+1} = \frac{na_n + \arctg(a_n)}{n+1} \dots\dots$	1pts
2.	Observing monotonicity and finding limit.....	1pts
3.	Approximating with Taylor for $\arctg(x)$	1pts
4.	Gettig identity with Taylor $a_{n+1} = a_n \left(1 - \frac{a_n^2}{3(n+1)} + \frac{o(a_n^2)}{n+1} \right) \dots\dots$	1pts
5.	Gettig identity $\frac{1}{a_{n+1}^2} = \frac{1}{a_n^2} + \frac{2}{3(n+1)} + \frac{o(1)}{n+1} \dots\dots$	2pts
6.	Idea of using Stolz-Cesaro theorem.....	1pts
7.	Combing (3) and (4)(Using Stolz-Cesaro theorem in needed form).....	2pts

Problem3 Let $a_1, \dots, a_n, b_1, \dots, b_n$ be unit vectors in R^n . Show that there exist numbers $\varepsilon_i = \pm 1$ so that the $n \times n$ matrix C formed by the column vectors $c_i = a_i + \varepsilon_i b_i$ satisfies

$$|\det C| \leq 2^{\frac{n}{2}}.$$

Author: Karen Keryan,
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Solution

By Hadamard's inequality

$$\begin{aligned} |\det C|^2 &\leq \prod_{i=1}^n \|c_i\|^2 = \prod_{i=1}^n (a_i + \varepsilon_i b_i)(a_i + \varepsilon_i b_i) = \prod_{i=1}^n (\|a_i\|^2 + 2\varepsilon_i a_i b_i + \|b_i\|^2) \\ &= 2^n \prod_{i=1}^n (1 + \varepsilon_i a_i b_i) \end{aligned} \quad (1)$$

Denote $x_i = a_i \cdot b_i$ and $A(\varepsilon, x, n) = \prod_{i=1}^n (1 + \varepsilon_i x_i)$. Let us show by induction on n that

$$\sum_{\varepsilon_i = \pm 1} A(\varepsilon, x, n) = 2^n.$$

Obviously it holds for $n = 1$. By induction hypothesis we have

$$\begin{aligned} \sum_{\varepsilon: \varepsilon_1 = \pm 1, \dots, \varepsilon_n = \pm 1} A(\varepsilon, x, n) &= \sum_{\varepsilon: \varepsilon_1 = \pm 1, \dots, \varepsilon_n = \pm 1} A(\varepsilon, x, n-1)(1 + \varepsilon_n x_n) \\ &= (1 + x_n) \sum_{\varepsilon: \varepsilon_1 = \pm 1, \dots, \varepsilon_{n-1} = \pm 1} A(\varepsilon, x, n-1) \\ &\quad + (1 - x_n) \sum_{\varepsilon: \varepsilon_1 = \pm 1, \dots, \varepsilon_{n-1} = \pm 1} A(\varepsilon, x, n-1) = 2 \sum_{\varepsilon: \varepsilon_1 = \pm 1, \dots, \varepsilon_{n-1} = \pm 1} A(\varepsilon, x, n-1) \\ &= 2^n. \end{aligned}$$

Since the number of summands in (2) is 2^n , we get that there exist $\varepsilon_i = \pm 1$ so that $A(\varepsilon, x, n) \leq 1$. By we have $|\det C| \leq 2^{\frac{n}{2}}$.

Problem 3(Cat A) Marking Scheme

No	Steps	Points
1.	Idea of using Hadamard inequality.....	2pts
2.	Using Hadamad ineq and getting $ \det C ^2 \leq 2^n \prod_{i=1}^n (1 + \varepsilon_i a_i b_i)$ not additive with (1)..	4pts
3.	Claiming that $\sum_{\varepsilon_i = \pm 1} A(\varepsilon, x, n) = 2^n$	2pts
4.	Proof of (3)	2pts
5.	Noting that $A(\varepsilon, x, n) \leq 1$ and using it to show our inequality.....	2pts

Problem 4 Consider ODE system in polar coordinates

$$\begin{cases} \dot{r} = f\left(r\left(a + r \sin \frac{1}{r}\right)\right), \\ \dot{\varphi} = g\left(r\left(a + r \sin \frac{1}{r}\right)\right), \end{cases}$$

The functions f, g are differentiable, and $f(x) > 0$ when $x > 0$: If $r = 0$ we suppose that $\sin \frac{1}{r}$ is be equal to zero.

The point $(0,0)$ is a unique singular point of the system. We call the point stable if for any $\varepsilon > 0$ there exists $\delta > 0$ such that every solution with initial data in the δ –neighbourhood of zero never leaves the ε – neighborhood of zero. The point is called asymptotically stable if it is stable, and, in addition, every solution with initial data in δ – neighborhood of zero tends to zero.

Under what $a \in R$ the point $(0,0)$ is stable? Asymptotically stable? Unstable?

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Answer: Asymptotically stable under $a < 0$, just stable when $a = 0$ unstable when $a > 0$:

Solution: If $(0,0)$ is singular, it means $f(0) = 0$. Equation does not have φ in the right side, therefore rotation of the plane will move solutions to another solutions. Since solutions can't intersect, that means if $(r(t), \varphi(t))$ is a solution, then $r(t)$ is monotonic. Therefore, $r(t)$

either tends to constant or grow indefinitely. If it tends to constant, it must be such constants that

$$r = f\left(r\left(a + r \sin \frac{1}{r}\right)\right) = 0. \text{ So either } r = 0, \text{ or } \left(a + r \cdot \sin \frac{1}{r}\right) = 0.$$

Note that we also have solutions $r = \text{const} = c_{a,n}$, where $c_{a,n}$ – roots of

$$\left(a + r \cdot \sin \frac{1}{r}\right) = 0$$

and they are present for any a . Since solutions can't intersect, $r(t)$ from the neighborhood of zero can't grow indefinitely, it have to tend either to 0 or $c_{a,n}$.

When $a > 0$ $\dot{r} > 0$ near zero, so solutions $r(t)$ grow, they tend to least $c_{a,n} > 0$. Unstable case.

When $a < 0$ $\dot{r} > 0$ near zero, so solutions $r(t)$ decrease, tend to 0. Asymptotically stable case.

When $a = 0$ $c_{0,n}$ are $\frac{1}{\pi n}$. If solution is in ring with radii $\frac{1}{\pi n}, \frac{1}{\pi(n+1)}$ it will remain in that ring.

So it's stable case, but not asymptotically stable.

Problem 4 (Cat A) Marking Scheme

No	Steps	Points
1.	Getting fixed point $f(0) = 0$	1pts
2.	Invariance of solutions under rotation through zero	2pts
3.	Monotonicity of $r(t)$, $r(t) \rightarrow \infty$ or $r(t) \rightarrow 0$	1pts
3.	Getting $r = 0$ and $\left(a + r \cdot \sin \frac{1}{r}\right) = 0$	1pts
4.	Solutions of type $r = \text{const} = c_{a,n}$ analyzing solutions (they are not intersect etc.)	2pts
5.	Case $a > 0$ and case $a < 0$	2pts
6.	Case $a = 0$	1pts

Problem 5

Let A and B be two strictly positive definite $n \times n$ matrices. Show that

$$\det(tA + (1-t)B) \geq (\det A)^t (\det B)^{1-t}$$

for any $t \in [0,1]$.

Author: Marcin Zygmunt, University of Silesia, Poland

Solution

Lemma

Let $M > 0$ be a positive definite matrix.

Then

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-(Mx,x)} dx = \int_{-\infty}^{+\infty} dx_1 \dots \int_{-\infty}^{+\infty} e^{-(Mx,x)} dx_n = \sqrt{\frac{(2\pi)^n}{\det M}}$$

Where $x = (x_1, \dots, x_n) \in \mathbb{R}_+^n$.

Proof of lemma: If $\det M = 0$ then both sides equal infinity, so the equality holds. Assume now that $\det M > 0$. In this case there exist matrices: an orthogonal U and a diagonal $\Lambda =$

$\begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$ with $\lambda_1, \dots, \lambda_n > 0$, such that $M = U\Lambda U^{-1}$. Substituting $y = Ux$ in the integral leads to

$$\begin{aligned} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-(Mx,x)} dx &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-(\Lambda y,y)} |\det U^{-1}| dy = \int_{-\infty}^{+\infty} e^{-\lambda_1 y_1^2} dy_1 \cdots \int_{-\infty}^{+\infty} e^{-\lambda_n y_n^2} dy_n \\ &= \frac{\sqrt{2\pi}}{\sqrt{\lambda_1}} \frac{\sqrt{2\pi}}{\sqrt{\lambda_2}} \cdots \frac{\sqrt{2\pi}}{\sqrt{\lambda_n}} = \sqrt{\frac{(2\pi)^n}{\det M}}. \blacksquare \end{aligned}$$

If $t = 0$ or $t = 1$ the inequality in question becomes a trivial equality. In case $t \in (0,1)$ we'll use Hölder inequality

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x)G(x)dx \leq \left(\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x)^p dx \right)^{\frac{1}{p}} \left(\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} G(x)^q dx \right)^{\frac{1}{q}},$$

Where p, q satisfy $\frac{1}{p} + \frac{1}{q} = 1$, for $t = \frac{1}{p}$ and functions

$$F(x) = e^{-(tAx,x)} \text{ and } G(x) = e^{-((1-t)Bx,x)}.$$

We have

$$F(x)G(x) = e^{-((tA+(1-t)B)x,x)},$$

Hence applying the Hölder inequality and the above lemma gives

$$\begin{aligned} \sqrt{\frac{(2\pi)^n}{\det(tA + (1-t)B)}} &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x)G(x)dx \leq \left(\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x)^t dx \right)^{\frac{1}{t}} \left(\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} G(x)^{1-t} dx \right)^{\frac{1}{1-t}} \\ &= \left(\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-(Ax,x)} dx \right)^t \left(\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-(Bx,x)} dx \right)^{1-t} \\ &= \left(\sqrt{\frac{(2\pi)^n}{\det(A)}} \right)^t \left(\sqrt{\frac{(2\pi)^n}{\det(B)}} \right)^{1-t} = \sqrt{\frac{(2\pi)^n}{(\det A)^t (\det A)^{1-t}}}. \end{aligned}$$

The inequality in question follows directly.

Problem 5 (Cat A) Marking Scheme

Nº	Steps	Points
1.	Claiming unitarily diagonalizability of strictly positive definite matrix.....	2pts
2.	Claiming lemma	2pts
3.	Proving lemma not additive with (1).....	4pts
3.	Claiming Hölder's integral ineq or other useful ineq.....	1pts
4.	Using (2) and (3) and conclusion of result	2pts

Problem 6 Let f be continuous nonnegative real valued function on $[0, \pi]$. Prove that there exists $\alpha \in (0, \pi)$ such that

$$\int_0^\pi f(x) dx \int_0^{\alpha^2} x \cdot \sin(x) dx = \int_0^\alpha f(\sqrt{x}) d(x)$$

Solution

Letting

$$h(t) = g(t) \int_0^\pi f(x) dx - f(t) \int_0^\pi \sin(x) dx$$

Let

$$u(s) = \int_0^\pi H(x) dx = \int_0^s \int_0^x h(t) dt dx$$

then u is a continuous and differentiable function in $[0, 1]$ such that

$$u'(0) = H(0) = \int_0^0 h(t) dt = 0$$

and

$$u'(1) = H(1) = \int_0^\pi h(t) dt = \int_0^\pi \sin(t) dt \int_0^\pi f(x) dx - \int_0^\pi f(t) dt \int_0^\pi \sin(x) dx = 0$$

and by *Flett's Theorem* there exists

$$c \in (0, 1) \text{ such that } u'(c) = H(c) = \frac{u(c) - u(0)}{c - 0} = \frac{u(c)}{c}.$$

Hence

$$\int_0^c t \cdot h(t) dt = \int_0^c t d(H(t)) = [tH(t)]_0^c - \int_0^c H(t) dt = cH(c) - u(c) = 0.$$

then

$$\begin{aligned} \int_0^c t \cdot h(t) dt &= \int_0^c t \cdot \sin(t) dt \int_0^\pi f(x) dx - \int_0^c t \cdot f(t) dt \int_0^\pi \sin(x) dx = 0. \\ \int_0^c t \cdot \sin(t) dt \int_0^\pi f(x) dx &= \int_0^c t \cdot f(t) dt \int_0^\pi \sin(x) dx \end{aligned}$$

Note that $\int_0^\pi \sin(x) dx = 2$

$$\int_0^c t \cdot \sin(t) dt \int_0^\pi f(x) dx = 2 \int_0^c t \cdot f(t) dt = \int_0^c f(t) d(t^2) = \int_0^c f(t) d(t^2)$$

After changing $\alpha^2 = (\sqrt{c})^2$ and substitution $t^2 = x$ we get

$$\int_0^{\alpha^2} x \cdot \sin(x) dx \int_0^\pi f(x) dx = \int_0^\alpha f(\sqrt{x}) d(x)$$

Problem 6(Cat A) Marking Scheme

N ^o	Steps	Points
1.	Considering function $h(t)$	2pts
2.	Considering function $u(s)$	2pts
3.	Calculating endpoint values for i.e. for $u'(0) = u'(1) = 0$	1pts
4.	Using of Flett's theorem finding $u'(c) = \frac{u(c)}{c}$	1pts
5.	Further calculations after stating (4)	2pts
6.	Using $\int_0^\pi \sin(x) dx = 2$ changing.....	1pts
7.	Conclusion of solution substitutions.....	1pts

Individual Olympiad Category B

Problem 1

Suppose we are given 2023 displays and 2023 remotes, but we don't know which display corresponds to which remote. In one operation, we can specify an arbitrary set of remotes, and all of them will be switched from **power off** to **power on** or vice versa simultaneously. We will then see which displays come on (initially they are all off). Find the least number of operations to determine which remote corresponds to which display?

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Solution Same as Problem1 Category A.

Problem 2

Let $a > b > 0$ and the ellipse ε is given by:

$$\varepsilon: \left| \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \right.$$

Let F be one of the foci of ε . Let A and B be points on the ellipse such that F lies on the segment AB . Let C be the intersecting point of the tangents to ε through the points A and B . Prove that:

- a) $CF \perp AB$;
- b) C lies on a directrix of the ellipse.

Authors: Asen Bozhilov and Stoyan Apostolov, Sofia University, Bulgaria

Solution

It is convenient to reformulate the problems as follows: Let $F(c, 0)$ be a focus and $l: x = \frac{a^2}{c}$ be the corresponding directrix, where $c = \sqrt{a^2 - b^2}$. Let $C(x_0, y_0)$ be a point on l of ε and let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the points on ε such that the tangents through A and B intersect at point C . Then we go on to show that F lies on AB and $CF \perp AB$.

In order to do this, recall that the tangent passing through A is given by $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$, and the tangent through B : $\frac{xx_2}{a^2} + \frac{yy_2}{b^2} = 1$. Now observe that A and B lie on the line $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$ (the so called *polar* of the point C). Indeed, since C lies on the tangent through A we have $\frac{x_1x_0}{a^2} + \frac{y_1y_0}{b^2} = 1$ and similarly for B . In order to show that F lies on AB it suffices to show that F lies on the same line. But since $x_0 = \frac{a^2}{c}$ the line through A and B is just $\frac{x}{c} + \frac{yy_0}{b^2} = 1$ which is clearly satisfied by the point $F(c; 0)$: It remains

to notice that the vector FC equals $\left(\frac{a^2}{c} - c, y_0\right)$. Hence it is collinear with the normal vector $\left(\frac{1}{c}, \frac{y_0}{b^2}\right)$ to the line AB , which shows that $AB \perp CF$.

Problem 2(Cat B) Marking Scheme

№	Steps	Points
1.	For fully written equation of directrix i.e for noting $l: x = \frac{a^2}{c}$	1pts
2.	Substituted tangent line equation of points A and B $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	4pts
3.	Observing equation of polar of C i.e line AB.....	2pts
4.	Coordinates of vector FC and collinearity to the normal vector of line AB	3pts

Problem3

Let A, B and C be 2023×2023 matrices with real entries such that:

$$AB + BC + CA = 2023I_{2023}$$

Find all possible values of $rank(A^2 + B^2 + C^2)$

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Solution

If the matrix $A^2 + B^2 + C^2$ is singular then there exists nonzero vector $\vec{x}$ such that

$$(A^2 + B^2 + C^2)\vec{x} = \vec{0}$$

Multiplying by $\vec{x}^T$ from the left we get that

$$\vec{0} = \vec{x}^T (A^2 + B^2 + C^2) \vec{x} = \vec{x}^T A^T \cdot A \vec{x} + \vec{x}^T B^T \cdot B \vec{x} + \vec{x}^T C^T \cdot C \vec{x} = \|y_1\|^2 + \|y_2\|^2 + \|y_3\|^2$$

where $y_1 = A\vec{x}, y_2 = B\vec{x}, y_3 = C\vec{x}$ and note that $\|y_1\|^2, \|y_2\|^2, \|y_3\|^2 \geq 0$ we get that

$$y_1 = A\vec{x} = y_2 = B\vec{x} = y_3 = C\vec{x} = \vec{0}$$

Then multiplying by $\vec{x}$ from right the given equation $AB + BC + CA = 2023I_{2023}$ we get

$$2023\vec{x} = (AB + BC + CA)\vec{x} = A(B\vec{x}) + B(C\vec{x}) + C(A\vec{x}) = \vec{0}$$

Hence $\vec{x} = \vec{0}$ which is contradiction so the matrix $A^2 + B^2 + C^2$ is nonsingular. Therefore only possibility is $rank(A^2 + B^2 + C^2) = n$.

Problem 3(Cat B) Marking Scheme

№	Steps	Points
1.	Using singularity of $A^2 + B^2 + C^2$ i.e. getting $(A^2 + B^2 + C^2)\vec{x} = \vec{0}$	1pts
2.	Getting $y_1 = A\vec{x} = y_2 = B\vec{x} = y_3 = C\vec{x} = \vec{0}$ with according calculations.....	5pts
2.	Getting contradiction $\vec{x} = \vec{0}$ using $AB + BC + CA = 2023I_{2023}$	3pts
3.	For stating $rank(A^2 + B^2 + C^2) = n$	1pts

Problem 4 Calculate limit

$$\lim_{n \rightarrow \infty} \sqrt[n]{\prod_{k=1}^n \Gamma\left(\frac{k}{n}\right)}$$

Where $\Gamma(x)$ -Gamma function.

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Solution

By using well known identity for Gamma function $\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)}$ we get

$$\Gamma\left(\frac{r}{n}\right)\Gamma\left(1-\frac{r}{n}\right) = \frac{\pi}{\sin\left(\frac{\pi r}{n}\right)} \quad r = 1, 2, 3, \dots, (n-1) \text{ equalities.}$$

$$\left[\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{2}{n}\right)\Gamma\left(\frac{3}{n}\right)\dots\Gamma\left(\frac{n-1}{n}\right)\right]^2 = \frac{\pi^{n-1}}{\sin\left(\frac{\pi}{n}\right)\sin\left(\frac{2\pi}{n}\right)\sin\left(\frac{3\pi}{n}\right)\dots\sin\frac{(n-1)\pi}{n}}$$

Consider,

$$x^n - 1 = \prod_{k=0}^{n-1} (x - \omega^k)$$

$$1 + x + x^2 + x^3 + \dots + x^{n-1} = \prod_{k=1}^{n-1} (x - \omega^k)$$

Let

$$x = 1; n = \prod_{k=1}^{n-1} (1 - \omega^k)$$

$$\text{And } \omega^k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n} = e^{i \frac{2\pi k}{n}} \quad 0 \leq k \leq n-1$$

Take modules on both sides,

$$\begin{aligned} |n| &= \prod_{k=1}^{n-1} |1 - \omega^k| \\ |1 - \omega^k| &= \left| 1 - \cos \frac{2\pi k}{n} - i \sin \frac{2\pi k}{n} \right| = \\ &= \sqrt{\left(1 - \cos \frac{2\pi k}{n}\right)^2 + \sin^2 \frac{2\pi k}{n}} = \sqrt{2 - 2\cos \frac{2\pi k}{n}} = \\ &= \sqrt{2\left(1 - \cos \frac{2\pi k}{n}\right)} = \sqrt{4\sin^2 \frac{\pi k}{n}} = 2\sin \frac{\pi k}{n} \end{aligned}$$

Hence,

$$n = 2^{n-1} \prod_{k=1}^{n-1} \sin\left(\frac{\pi k}{n}\right)$$

Or,

$$\begin{aligned} \prod_{k=1}^{n-1} \sin\left(\frac{\pi k}{n}\right) &= \frac{n}{2^{n-1}} \\ \left[\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{2}{n}\right)\Gamma\left(\frac{3}{n}\right)\dots\Gamma\left(\frac{n-1}{n}\right)\right]^2 &= \frac{\pi^{n-1}}{\frac{n}{2^{n-1}}} = \frac{2^{n-1}}{n} \pi^{n-1} \end{aligned}$$

Hence,

$$\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{2}{n}\right)\Gamma\left(\frac{3}{n}\right)\dots\Gamma\left(\frac{n-1}{n}\right) = \sqrt{\frac{2^{n-1}}{n}} \cdot \pi^{\frac{n-1}{2}} = \frac{(2\pi)^{\frac{n-1}{2}}}{n^{\frac{1}{2}}}$$

Then

$$\lim_{n \rightarrow \infty} \sqrt[n]{\prod_{k=1}^{n-1} \Gamma\left(\frac{k}{n}\right)} = \lim_{n \rightarrow \infty} \frac{(2\pi)^{\frac{n-1}{2n}}}{n^{\frac{1}{2n}}} = \sqrt{2\pi}$$

Problem 4 (Cat B) Marking Scheme

N ^o	Steps	Points
1.	Property $\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)}$	1pts
2.	Using of (1) i.e. identity like $\Gamma\left(\frac{r}{n}\right)\Gamma\left(1-\frac{r}{n}\right) = \frac{\pi}{\sin\left(\frac{\pi r}{n}\right)}$ $r = 1, 2, 3, \dots, (n-1)$	1pts
3.	For $\left[\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{2}{n}\right)\Gamma\left(\frac{3}{n}\right)\dots\Gamma\left(\frac{n-1}{n}\right)\right]^2 = \frac{\pi^{n-1}}{\sin\left(\frac{\pi}{n}\right)\sin\left(\frac{2\pi}{n}\right)\sin\left(\frac{3\pi}{n}\right)\dots\sin\frac{(n-1)\pi}{n}}$	3pts
3.	For $1+x+x^2+x^3+\dots+x^{n-1} = \prod_{k=1}^{n-1}(x-\omega^k)$	1pts
4.	Calculations and getting $\prod_{k=1}^{n-1} \sin\left(\frac{\pi k}{n}\right) = \frac{n}{2^{n-1}}$	3pts
5.	Last 2 rows of solution.....	1pts

Note: Here can be alternate solution using Riemann sum not additive with above points

N ^o	Steps	Points
1.	Writing Riemann sum	1pts
2.	Claiming that the given is equal to $\exp\left(\int_0^1 \ln(\Gamma(x))dx\right)$ not additive with (1).....	4pts
3.	Calculating integral $\exp\left(\int_0^1 \ln(\Gamma(x))dx\right)$ and final correct answer.....	6pts

Problem 5

Let $A \in M_n(\mathbb{R})$ and $A^t A^2 A^t = A + A^t$. Prove that

a) $\det(A) \geq 0$.

b) $\text{Tr}(A) \geq 0$. When does equality occur?

Authors: Asen Bozhilov and Stoyan Apostolov, Sofia University, Bulgaria

Solution

a) Transposing the equality gives $A(A^t)^2 A = A + A^t$, hence the matrices AA^t and $A^t A$ commute. Since both are real symmetric, they are diagonalizable by orthogonal matrices. But since they commute, they are simultaneously diagonalizable, i.e. there exists an orthogonal matrix U such that $UAA^t U^t = D_1$ and $UA^t A U^t = D_2$, where D_1 and D_2 are diagonal matrices. Observe that AA^t and $A^t A$ are positive semi-definite, hence the elements of D_1 and D_2 are nonnegative. Multiplying the original equation by U from the left and U^t from the right we obtain that $D_1 D_2 = U(A + A^t)U^t$,

hence $U^t D_1 D_2 U = A + A^t$: Thus $A + A^t$ is positive semi-definite. This shows that for any $v \in \mathbb{R}^n$ it holds that $((A + A^t)v, v) \geq 0$. Using the properties of the scalar product we obtain

$$(Av, v) + (A^t v, v) = (Av, v) + (v, Av) = 2(Av, v) \geq 0$$

For all $v \in \mathbb{R}^n$. This shows that all eigenvalues of A , i.e. all real characteristic roots of A , are nonnegative. On the other hand, since A is real matrix, the nonreal characteristic roots of A come in conjugate pairs. Since $\det(A)$ is the product of all characteristic roots, the result follows.

b) Taking the trace of both sides on the equation gives

$$\text{Tr}(A^t A^2 A^t) = \text{Tr}(A + A^t)$$

Using basic properties of the trace we obtain $\text{Tr}(A^2 (A^2)^t) = 2\text{Tr}(A)$. Now recall the inequality $\text{Tr}(XX^t) \geq 0$ with equality only if $X = 0$. Applied with $X = A^2$ we obtain that $\text{Tr}(A) \geq 0$. Equality occurs when $A^2 = 0$. This implies that the characteristic roots of A are all 0. Moreover, the original equation now reduces to $A + A^t = 0$; i.e. A is an antisymmetric matrix. This implies that A is diagonalizable. And the diagonal matrix should be 0. Hence $A = 0$.

Problem 5 (Cat B) Marking Scheme

No	Steps	Points
1.	showing AA^t and $A^t A$ commute	1pts
2.	Claiming simultaneous diag of AA^t and $A^t A$	1pts
3.	Diagonal forms using positive definitivity of matrices AA^t and $A^t A$	1pts
4.	Claiming $D_1 D_2 = U(A + A^t)U^t$ positive definiteness of matrix $A + A^t$	1pts
5.	Showing $2(Av, v) \geq 0$	1pts
6.	Stating eigenvalues of A are characteristic roots and finish part (b)	1pts
7.	Taking trace and getting $\text{Tr}(A^t A^2 A^t) = \text{Tr}(A + A^t)$	1pts
8.	Equality case when $A^2 = 0$	2pts
9.	Getting that A can be only 0 using diag	1pts

Problem 6

Let $f: [0,1] \rightarrow [0,1]$ be a continuous function. Assume that there exists $a \in [0; 1]$ such that $f(a) < a < f(f(a))$.

Prove that there exists $z \in [0,1]$ such that $f(z) \neq z$ and $f(f(z)) = z$.

Authors: Asen Bozhilov and Stoyan Apostolov, Sofia University, Bulgaria

Solution

According to the Intermediate value theorem (IVT), there exists $b \in (f(a), a)$ such that $f(b) = a$. Observe that for $g(x) = f(f(x)) - x$ we have $g(b) = f(a) - b < 0$ and $g(0) \geq 0$, so that IVT implies that there exists $c \in [0, b)$ with $g(c) = 0$, i.e. $f(f(c)) = c$. If the set of fixed points of f denote it $\Omega = \{x | f(x) = x\}$ has empty intersection with $[0, b]$, we are done. Now assume that $\Omega \cap [0, b] \neq \emptyset$. Let $\alpha = \sup(\Omega \cap [0, b])$. Since f is continuous Ω is closed, hence $\alpha \in \Omega \cap [0, b]$. Since $f(b) \neq b$ we obtain $\alpha < b$. Since $f(\alpha) = \alpha$ and $f(b) = a > b$ IVT implies that there exists $c \in (\alpha, b)$ such that $f(c) = b$. Thus $g(c) = f(f(c)) - c = a - c > 0$. Moreover $g(b) = f(f(b)) - b = f(a) - b < 0$. Thus there exists $z \in (c, b)$ such that $g(z) = 0$, i.e. $f(f(z)) = z$. Since $\alpha < z < b$ we obtain that $f(z) \neq z$.

Problem 6(Cat B) Marking Scheme

Nº	Steps	Points
1.	Using IVT and getting b with $f(b) = a$	1pts
2.	Observation of $g(x) = f(f(x)) - x$ and finding c with $f(f(c)) = c$	3pts
3.	Considering Ω and using $\alpha = \sup(\Omega \cap [0, b])$	2pts
4.	Finding c with $f(c) = b$ and $c \in (\alpha, b)$	2pts

Team Olympiad

Problem 1

Alice and Bob play a game with a bar of chocolate made of 60 pieces, in a 20×23 rectangle. The first player Alice breaks off a part of the chocolate bar along the grooves dividing the pieces, and discards (or eats) the part she broke off. Then the second player Bob breaks off a part of the remaining part and discards his part. The game continues until one piece is left. The winner is the one who leaves the other with the single piece (i.e. is the last to move). Which person has a perfect winning strategy?

Authors: Pirmyrat Gurbanov and Murat Chashemov,
International University for the Humanities and Development, Turkmenistan

Solution 1. Let's solve the problem in reverse order: The player with 1×1 is an obvious sure losing situation for the one who is stuck with it; it can not be broken, so the game is over. $1 \times n$ bars are all sure winners (n greater than 1), because the breaker can leave the other player with the sure loser situation, 1×1 . Note that 2×2 is a sure losing situation, because the breaker must end up with a 1×2 bar, which is a sure winning situation for the other player. Now, we can say that $2 \times n$ bars (with $n > 2$) are sure winning situation, because we can land the other player with the sure losing, 2×2 . And so forth. We note that:

(i) If $a \times b$ is a sure losing situation, then $a \times c$ (with $c > b$) is a sure winning situation, because the person who breaks $a \times c$ should leave the other player with $a \times b$. For example, because we have shown that 2×2 is a sure losing situation, then 2×4 , 2×5 , 2×6 , etc. are all sure winning situations.

(ii) $a \times b$ is a sure losing situation only when all the possible moves from it are sure Winning situation for the other player. For example, 1×4 , 2×4 , 3×4 , and by symmetry 4×3 , 4×2 , 4×1 are all sure losing situations, as we have shown above, hence 4×4 must be a sure losing situation.

Let's do this method of counting until we reach 20×23 bar there we write winning situations for player with bolded font

(1×1) **(1×2)** (1×3) (1×4) (1×5) ... (1×23)
 (2×1) (2×2) (2×3) **(2×4)** (2×5) ... (2×23)
 (3×1) (3×2) (3×3) (3×4) **(3×5)** ... (3×23)
 (4×1) (4×2) (4×3) (4×4) (4×5) ... (4×23)
 (5×1) (5×2) (5×3) (5×4) (5×5) ... (5×23)
.
.
.
 (23×1) (23×2) (23×3) (23×4) **(23×5)** ... (23×23)

So we start with **(20×23)** bar and it's winning situation for first player because he make first move.

Solution 2.

Claim: Any square $(n \times n)$ bar is losing situation for player.

Proof: If the some player moves with square bar in the next turn other player can leave that player again with square of smaller size $n > s$ ($s \times s$) by continuing inductively and note that the (1×1) is least square bar and losing situation so that player will lose■

First player brokes bar such that second player left with (20×20) bar by our claim first player wins.

Problem 1(Team) Marking Scheme (Solution1)

No	Steps	Points
1.	Idea of going backwards.....	3pts
2.	Observing that $1 \times n$ is winning situation.....	3pts
3.	Constructing the table for $n > 23$	3pts
4.	Finding winner.....	1pts

Problem 1(Team) Marking Scheme (Solution2)

No	Steps	Points
1.	Claiming that $n \times n$ is losing situation.....	3pts
2.	Showing claim (1).....	4pts
3.	Constructing strategy to player.....	3pts

Problem 2

Find all functions $f: R \rightarrow R$, such that for all $x, y \in R$ the equality

$$(x + y)(f(x) - f(y)) = (x - y)f(x + y)$$

holds true.

Author: Yagub Aliyev, ADA University, Azerbaijan

Solution

First substitute $x = t, y = 1$ to get

$$(t + 1)(f(t) - f(1)) = (t - 1)f(t + 1).$$

Similarly substitute $x = -1, y = t + 1$ to get

$$t(f(-1) - f(t + 1)) = (-t - 2)f(t).$$

Multiply the first equality by t , the second one by $t - 1$, and add them to obtain an equation without $f(t + 1)$. So we get

$$f(t) = At + B,$$

where $A = -\frac{f(1)+f(-1)}{3}, B = \frac{f(1)-f(-1)}{3}$.

If we check this in the given equation then we get

$$(x + y)(Ax - Ay) = (x - y)(A(x + y) + B). \text{ By simplifying, we obtain } B = 0.$$

So, the answer is $f(x) = Ax$, where A is an arbitrary constant.

Problem 2(Team) Marking Scheme

No	Steps	Points
1.	Putting $x = t, y = 1$ getting identity.....	3pts
2.	Putting $x = -1, y = t + 1$ getting identity.....	3pts
3.	Getting that f is linear $f(t) = At + B$	2pts
4.	Finding solution $f(x) = Ax$	2pts

Problem 3

Let a, b, c be fixed positive real number, such that there is a triangle with sides of length a^n, b^n, c^n for any positive integer n . Show that all these triangles are isosceles (i.e. two of three numbers a, b, c are equal).

Author: Marcin Zygmunt, University of Silesia, Poland

Solution Without loss of generality we can assume that $0 < a \leq b \leq c$. We have

$$c^n \leq a^n + b^n.$$

so

$$1 \leq \left(\frac{a}{c}\right)^n + \left(\frac{b}{c}\right)^n. \quad (1)$$

If $c > \max\{a, b\}$ then the right-hand side of (1) tends to 0 as n tends to infinity, which leads to the contradiction. Hence $c = \max\{a, b\} = b$ and all triangles with sides $a^n, b^n = c^n$ are isosceles.

Problem 3(Team) Marking Scheme

Nº	Steps	Points
1.	Getting the inequality (1) for all natural n	5pts
2.	Passing to the limit n tends to infinity and getting contradictin.....	5pts

Problem 4

Series

$$\sum_{n=1}^{\infty} x_n = 1 - 1 + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \cdots + \underbrace{\frac{1}{2^n} - \frac{1}{2^n} + \cdots + \frac{1}{2^n} - \frac{1}{2^n}}_{2^{n+1} \text{ summands}} + \cdots$$

is given

- 1) Find the sum of given series.
- 2) For bijection $\pi: \mathbb{N} \rightarrow \mathbb{N}$, defined below

$$\pi(n) = \begin{cases} 2k, & \text{if } n = 3k; \\ 4k - 1, & \text{if } n = 3k - 1; \\ 4k - 3, & \text{if } n = 3k - 2. \end{cases}$$

Find sum $\sum_{n=1}^{\infty} x_{\pi(n)}$.

- 3) Construct the bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}$ such that $\sum_{n=1}^{\infty} x_{\sigma(n)} = 2$

Author: Leonid Genze, Tomsk State University, Russian Federation

Solution

a) Let $S_N = \sum_{k=1}^N x_k$ and $S = \lim_{N \rightarrow \infty} S_N$. It's easy to see that $S_{2N} = 0$ for each $N \in \mathbb{N}$ and $S_{2N+1} = \frac{1}{2^{n(N)}}$ for some $n(N) \in \mathbb{N}$, which $n(N) \rightarrow +\infty$ when $N \rightarrow +\infty$. Then $\lim_{N \rightarrow \infty} S_{2N+1} = 0$ and $S = 0$.

b) $\tilde{S}_{3N+1} = \sum_{k=1}^{3N+1} x_{\pi(k)} = 1 + \underbrace{\frac{1}{2} - 1 + \frac{1}{2}}_{=0} + \underbrace{\frac{1}{4} - \frac{1}{2} + \frac{1}{4}}_{=0} + \underbrace{\frac{1}{8} - \frac{1}{4} + \frac{1}{8}}_{=0} + \cdots = 1$ for each $N \in \mathbb{N}$.

$\tilde{S}_{3N+1}, \tilde{S}_{3N-1} \in \left(\tilde{S}_{3N+1} - \frac{1}{2^{n(N)}}, \tilde{S}_{3N+1} + \frac{1}{2^{n(N)}} \right)$, where $n(N) \rightarrow +\infty$ for $N \rightarrow +\infty$

Hence, $\lim_{N \rightarrow \infty} \tilde{S}_{3N} = \lim_{N \rightarrow \infty} \tilde{S}_{3N-1} = \lim_{N \rightarrow \infty} \tilde{S}_{3N+1} = 1$, finally we get $\sum_{n=1}^{\infty} x_{\pi(n)} = 1$.

c) For constructing σ note that, $\tilde{S}_{3N} = U_N + V_{2N}$ where $U_N = \sum_{k=1}^N x_{2k}$, $V_N = \sum_{k=1}^N x_{2k-1}$. From a) we get that $\lim_{N \rightarrow \infty} (U_N + V_N) = 0$, also from b) we get that $\lim_{N \rightarrow \infty} (U_N + V_{2N}) = 1$.

Thus,

$$U_N + V_{4N} = (U_N + V_{2N}) - (V_{2N} + U_{2N}) + (U_{2N} - V_{4N}) \rightarrow 1 - 0 + 1 = 2.$$

Therefore, σ can be chosen as below:

For each negative term of sum (from U_{2N}), we take 4 positive terms of sum (from V_{4N}), without changing the order of positive and negative terms. Example,

$$\sigma(n) = \begin{cases} 2k, & n = 5k; \\ 8k - 1, & n = 5k - 1; \\ 8k - 3, & n = 5k - 2; \\ 8k - 5, & n = 5k - 3; \\ 8k - 7, & n = 5k - 4. \end{cases}$$

In this case we get

$$\sum_{n=1}^{\infty} x_{\sigma(n)} = 1 + \underbrace{\frac{1}{2} + \frac{1}{2}}_{=2} + \frac{1}{4} - 1 + \underbrace{\frac{1}{4} + \frac{1}{4} + \frac{1}{4}}_{=0} + \frac{1}{8} - \frac{1}{2} + \underbrace{\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}}_{=0} - \frac{1}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} - \frac{1}{4} + \dots$$

Another case

$$\begin{aligned} \sum_{n=1}^{\infty} x_{\sigma_1(n)} &= 1 + \underbrace{\frac{1}{2} + \frac{1}{2}}_{=2} + \underbrace{\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}}_{=0} - 1 + \underbrace{\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}}_{=0} - \frac{1}{2} + \underbrace{\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}}_{=0} - \frac{1}{2} \\ &+ \underbrace{\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16}}_{=0} - \frac{1}{4} + \dots \end{aligned}$$

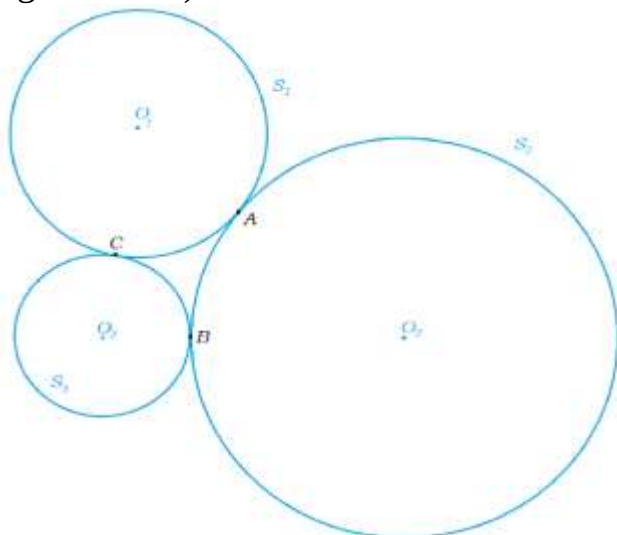
Problem 4(Team) Marking Scheme

No	Steps	Points
1.	Calculating part a)	2pts
2.	Correct example for b).....	4pts
3.	Correct example for c).....	4pts
4.	Idea of interchanging places of terms in sum not additive with (1), (2), (3).....	1pts
5.	Theorem about the sum of convergent alternating series not additive with (1), (2), (3).....	4pts

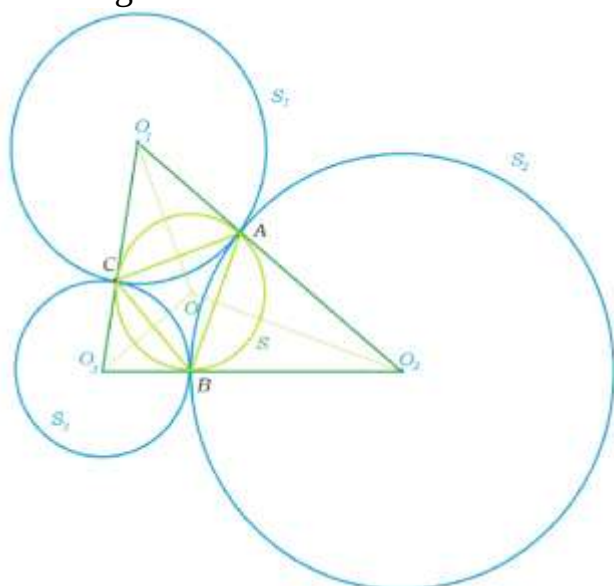
Problem 5 Let A, B, C be three given points in the plane, which form an acute-angled triangle. Construct three circles S_1, S_2, S_3 such that circles S_1 and S_2 are tangent at point A , circles S_2 and S_3 at point B , and circles S_1 and S_3 at point C .

Author: Marcin Zygmunt, University of Silesia, Poland

Solution Let us suppose that O_1, O_2, O_3 be centers of the respective circles S_1, S_2, S_3 (see figure below).



The segments O_1O_2 , O_2O_3 and O_3O_1 pass through the corresponding points A , B and C . The symmetrical of the segment AB is also the bisector of angle $\angle AO_1B$, i.e. of the angle $\angle O_2O_1O_3$. The same applies to the other segments. Therefore, the circle circumscribed on the triangle ΔABC is the circle inscribed in the triangle $\Delta O_1O_2O_3$ (see figure below).



This means that all three segments O_1O_2 , O_2O_3 and O_3O_1 are tangent to the circle S . Therefore, in order to solve the problem, we have first pass the perpendiculars of the sides of triangle ΔABC . Let us denote their intersection by O . Then we draw the lines OA , OB and OC . Finally, we construct perpendiculars to them at points A , B and C . The points of intersection of these perpendiculars will be the centers of the circles. The problem has a solution for all points A , B and C being vertices of an acute-angled triangle.

Problem 5(Team) Marking Scheme

No	Steps	Points
1.	Considering incircle of $\Delta O_1O_2O_3$ and tangency of all O_iO_j to circle S	4pts
2.	Drawing OA, OB, OC and taking ependiculas at points A,B,C	3pts
3.	Showing that the intersection of perpendiculars are the centers of the circles.....	3pts

Problem 6 Compute the sum

$$\sum_{k=1}^n \tan^2\left(\frac{\pi k}{2n+1}\right).$$

Author: Karen Keryan,
American University of Armenia and Yerevan State University, Armenia

Solution

Consider $P_n(x) = (x^2 + 1)^{n+1/2} \sin((2n+1) \arctan(x))$. Let's show that $P_n(x)$ is a polynomial. Notice that

$$\begin{aligned} 2i \sin(n\alpha) &= e^{in\alpha} - e^{-in\alpha} = (\cos(\alpha) + i \sin(\alpha))^n - (\cos(\alpha) - i \sin(\alpha))^n \\ &= \sum_{k=0}^n \binom{n}{k} \cos^k \alpha \sin^{n-k} \alpha (i^{n-k} - (-i)^{n-k}) \\ &= \sum_{k=0}^n \binom{n}{k} \cos^k \alpha \sin^{n-k} \alpha (e^{\frac{\pi i(n-k)}{2}} - e^{-\frac{\pi i(n-k)}{2}}) = 2i \sin^n \alpha \sum_{k=0}^n \binom{n}{k} \cot^k \alpha \sin\left(\frac{\pi(n-k)}{2}\right). \end{aligned}$$

Hence

$$\begin{aligned} P_n(x) &= (x^2 + 1)^{n+1/2} \cdot \sin^{2n+1}(\arctan x) \sum_{k=0}^{2n+1} \binom{2n+1}{k} \cot^k(\arctan x) \sin\left(\frac{\pi(2n+1-k)}{2}\right) \\ &= x^{2n+1} \sum_{k=0}^{2n+1} \binom{2n+1}{k} \frac{1}{x^k} \sin\left(\frac{\pi(2n+1-k)}{2}\right) \\ &= (-1)^n x^{2n+1} + (-1)^{n-1} \binom{2n+1}{2} x^{2n-1} + \dots + \binom{2n+1}{2n} x. \end{aligned}$$

Denote $x_k = \frac{\tan \pi k}{2n+1}$, $k = -n, \dots, n$. Clearly $x_{-n}, \dots, x_n$ are roots of P_n and as $\deg P_n = 2n+1$, it has no other roots. By Vieta's theorem $\sum_{i=-n}^n x_i = 0$ and $\sum_{-n \leq i < j \leq n} x_i x_j = -\binom{2n+1}{2}$.

Hence

$$\sum_{k=1}^n \tan^2\left(\frac{\pi k}{2n+1}\right) = \sum_{k=1}^n x_k^2 = \frac{1}{2} \sum_{k=-n}^n x_k^2 = \frac{1}{2} \left(\left(\sum_{i=-n}^n x_i \right)^2 - 2 \sum_{-n \leq i < j \leq n} x_i x_j \right) = \binom{2n+1}{2} = n(2n+1).$$

Problem 6(Team) Marking Scheme

Nº	Steps	Points
1.	Getting $2i \sin(n\alpha) = 2i \sin^n \alpha \sum_{k=0}^n \binom{n}{k} \cot^k \alpha \sin\left(\frac{\pi(n-k)}{2}\right)$	4pts
2.	For $P_n(x) = (-1)^n x^{2n+1} + (-1)^{n-1} \binom{2n+1}{2} x^{2n-1} + \dots + \binom{2n+1}{2n} x$	3pts
3.	For finding roots of $P_n(x)$ and using Viet's theorem on $P_n(x)$	3pts
4.	Correctly calculated sum of series by using (3).....	1pts

Problem 7

Let A, B be $n \times n$ matrices such that $a_{ij} - b_{ij} = 1$ for all i, j . Find all natural numbers k such that the condition $A^k = 0$ implies that $\det B = 0$.

Author: Karen Keryan,
American University of Armenia and Yerevan State University, Armenia

Solution Let us prove that $A^k = 0$ implies that $\det B = 0$ iff $k \leq n - 1$.

Denote $C = \{1\}_{i,j=1}^n$. So we have $A - B = C$.

Notice that if $A^k = 0$ for some $k \leq n - 1$, then the eigenvalues of A are 0's and the Jordan blocks are of size not exceeding k . Therefore the Jordan form of A should consist of at least two Jordan blocks. Since the rank of such A is equal to n minus the number of Jordan blocks, we have that $\text{rk} A \leq n - 2$. Note that $\text{rk} C = 1$, hence $\text{rk} B = \text{rk}(A - C) \leq \text{rk} A + \text{rk} C \leq n - 1$. Hence $\det B = 0$.

In the case $k \geq n$ consider the matrix $A = \begin{bmatrix} 0 & -1 & 0 & \dots & 0 \\ 0 & 0 & -1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & -1 \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$. It is easy to verify that

$A^n = 0$, hence $A^k = 0$. Subtracting row i from row $i - 1$ for $i = 2, \dots, n$ and then expanding along the first column we get

$$\begin{aligned} \det(C - A) &= \begin{vmatrix} 1 & 2 & 1 & \dots & 1 \\ 1 & 1 & 2 & \dots & 1 \\ \dots & \dots & \dots & \dots & \dots \\ 1 & 1 & 1 & \dots & 2 \\ 1 & 1 & 1 & \dots & 1 \end{vmatrix} = \begin{vmatrix} 0 & 1 & -1 & 1 & \dots & 0 \\ 0 & 0 & 1 & -1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & 1 \\ 1 & 1 & 1 & 1 & \dots & 1 \end{vmatrix} \\ &= (-1)^{n+1} \begin{vmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & -1 \\ 0 & 0 & 0 & \dots & 1 \end{vmatrix} = (-1)^{n+1} \neq 0. \end{aligned}$$

Therefore taking $B = A - C$ we get that $\det B = (-1)^n \det(C - A) \neq 0$ and $A - B = C$

Problem 7(Team) Marking Scheme

No	Steps	Points
1.	Case $k \leq n - 1$ Finding Jordan form of matrix A and getting $\det B = 0$	5pts
2.	Case $k \geq n$ giving example and correct calculations.....	5pts

Problem 8 Denote the set of all nonnegative integer numbers with N_0 if $n \in N_0$, and $h(x)$ be a $4n + 3$ times continuously differentiable function on R show that there exists number a with

$$\prod_{k=0}^{4n+3} \sum_{t=0}^k \frac{h^t(a)}{t! (k-t)!} \geq 0.$$

Authors: Pirmyrat Gurbanov and Murat Chashemov,
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Solution

Consider function $f(x) = e^x h(x)$ then note that

$$f^{(k)}(x) = e^x \left(\sum_{t=0}^k C_n^k h^t(x) \right) = e^x \cdot k! \left(\sum_{t=0}^k \frac{h^t(x)}{t! (k-t)!} \right) \quad (1)$$

If we put $k = 0, 1, \dots, 4n + 3$ and multiply all of them we get

$$\prod_{k=0}^{4n+3} f^{(k)}(x) = \prod_{k=0}^{4n+3} e^x \cdot k! \cdot \left(\sum_{t=0}^k \frac{h^t(x)}{t!(k-t)!} \right) = e^{(4n+4)x} \cdot 1! \cdot 2! \cdot \dots \cdot (4n+3)! \cdot \prod_{k=0}^{4n+3} \sum_{t=0}^k \frac{h^t(a)}{t!(k-t)!}$$

Then our problem reduces to find $a \in \mathbf{R}$ with

$$\prod_{k=0}^{4n+3} f^{(k)}(a) \geq 0.$$

because $e^{(4n+4)x} \cdot 1! \cdot 2! \cdot \dots \cdot (4n+3)! > 0$

If at least one of the functions $f^{(k)}(x)$ vanishes at some point a , then we are done. Otherwise by the *Intermediate Value Theorem*, each $f^{(k)}(x)$ is either strictly positive or strictly negative on $\mathbf{R}$.

By replacing $f(x)$ by $-f(x)$ if necessary, we may assume the derivative $f^{(4n+2)}(x) > 0$. Since $4t+3$ is odd, by replacing $f(x)$ by $f(-x)$ if necessary, we may assume that also $f^{(4n+3)}(x) > 0$.

Since the number of factors, i.e. $4n+4$, is a multiple of 4, these substitutions do not change the sign of the whole product.

Lemma: if $g(x) \in C^2(0, +\infty)$, and $g'(x)$, $g''(x)$ are positive in $(a, +\infty)$ for $a \geq 0$ then there is a real number $b \geq a$ such that $g(x) > 0$ for all $x \in (b, +\infty)$.

Proof: Since g', g'' are positive then g' is positive and increasing. Thus, for $x > a$

$$g(x) = g(a) + \int_a^x g'(s) ds \geq g(a) + (x-a)g'(a).$$

and it follows that $g(x) > 0$ for $x > b$ with $b = \max(a, a - g(a)/g'(a))$.

Hence, by letting $a_{4n+3} = a_{4n+2} = 0$ and by using our claim for $g = f^{(4n+1)}(x)$ we find that $f^{(4n+1)}(x) > 0$ for all $x \in (a_{4n+1}, +\infty)$ for some $a_{4n+1} \geq a_{4n+2}$. Now we can apply our lemma for $g = f^{(4n)}(x)$ and therefore, by iterating this process we have that for any integer $0 \leq k \leq 4n+3$,

So, $f^{(k)}(x) > 0$ for all $x \in (a, +\infty)$ for some $a_0 \geq a_1 \geq \dots \geq a_{4n+3} = a_{4n+2} = 0$ and $f(x)$ finally for $a \in (a_0, +\infty)$

$$\prod_{k=0}^{4n+3} f^{(k)}(a) > 0.$$

Problem 8(Team) Marking Scheme

Nº	Steps	Points
1.	Considering function $f(x) = e^x h(x)$	2pts
2.	For identity $f^{(k)}(x) = e^x \cdot k! \left(\sum_{t=0}^k \frac{h^t(x)}{t!(k-t)!} \right)$	1pts
3.	For identity $\prod_{k=0}^{4n+3} f^{(k)}(x) = e^{(4n+4)x} \cdot 1! \cdot 2! \cdot \dots \cdot (4n+3)! \cdot \prod_{k=0}^{4n+3} \sum_{t=0}^k \frac{h^t(a)}{t!(k-t)!}$	1pts
4.	Reducing problem to $\prod_{k=0}^{4n+3} f^{(k)}(a) \geq 0$	1pts
5.	Sign invariancy i.e. part before lemma	1pts
6.	Claiming and using lemma without proof	1pts
7.	Proving and using lemma not additive with (6).....	5pts

Problem 9

Let $A \in M_n(\mathbb{C})$ and $AA^* + A^*A = 2I_n$, where I_n is the identity matrix of order n . Prove that

- if n is odd, then $\det(AA^* - I_n) = 0$;
- $|\det(A)| \leq 1$. When does equality occur?

Authors: Asen Bozhilov and Stoyan Apostolov, Sofia University, Bulgaria

Solution

Rewriting the equation in the form $AA^* = 2I_n - A^*A$ we obtain that AA^* and A^*A commute. Moreover both are Hermitian matrices, hence they are simultaneously diagonalizable by an unitary matrix. This means that for some unitary matrix U we have $UAA^*U^* = D_1$ and $UA^*AU^* = D_2$ where D_1 and D_2 are diagonal matrices. Multiplying the original equation by U from the left and U^* from the right we obtain

$$(1) \quad D_1 + D_2 = 2I_n$$

Recall that for any matrices X and Y , the characteristic polynomials of XY and YX are the same, hence their eigenvalues are the same (counted with multiplicities). Thus AA^* and A^*A have the same diagonal matrices up to rearrangement. Hence (1) implies that if λ is an eigenvalue of AA^* , so is $2 - \lambda$. We may group the eigenvalues in pairs $(\lambda, 2 - \lambda)$.

a) If n is odd, there must exist a pair consisting of two equal numbers, i.e. for some eigenvalue α it holds that $\alpha = 2 - \alpha$. Hence 1 is an eigenvalue of AA^* and the result follows.

b) The characteristic roots of AA^* are positive real numbers. The arguments above imply that the roots which are not equal to 1 could be arranged in pairs $\{\lambda_i, 2 - \lambda_i\}_{i=1}^k$ for some $k \leq \frac{n}{2}$ (in which case $n - 2k$ of the characteristic roots are equal to 1). In particular, all characteristic roots belong to $[0, 2]$. Since the determinant of a matrix is equal to the product of its characteristic roots we obtain

$$\det(AA^*) = 1^{n-2k} \prod_{i=1}^k \lambda_i(2 - \lambda_i) \leq 1^{n-2k} 1^k = 1$$

where we have used that if $a \in [0; 2] \setminus \{1\}$ holds that $a(1 - a) < 1$. Since $\det(AA^*) = \det(A) \det(A^*) = \det(A) \overline{\det(A)} = |\det(A)|^2$, the inequality follows. If equality holds, then all the characteristic roots of AA^* are equal to 1 which is equivalent (since AA^* is Hermitian) to $AA^* = I_n$. Clearly if this holds then $|\det(A)| = 1$. Thus equality occurs if and only if A is unitary.

Problem 9(Team) Marking Scheme

No	Steps	Points
1.	showing AA^* and A^*A commute	1pts
2.	Claiming simultaneous diag of AA^* and A^*A	1pts
3.	For getting (1) i.e. $D_1 + D_2 = 2I_n$	1pts
4.	Same characteristic polynomial for XY and YX and similarity	1pts
5.	Finding eigenvalue pairs of AA^* i.e. $(\lambda, 2 - \lambda)$	1pts
6.	Showing part (a) using above	1pts
7.	Showing that the characteristic roots belong to $[0,2]$	2pts
8.	Showing $\det(AA^*) \leq 1$	1pts
9.	Estimating equality case if and only if A is unitary	1pts

Problem 10

Let $f: Z_N \rightarrow S^1$ be a function such that for every $x, y \in Z_N$ we have

$$|f(x+y) - f(x)f(y)| < \varepsilon$$

where S^1 is the unit circle and ε is small constant. Prove that there is a function $g: Z_N \rightarrow S^1$ such that $g(x+y) = g(x)g(y)$ for all $x, y \in Z_N$ and $|g(x) - f(x)| < \varepsilon$.

Author: Omid Hatami, Institute of Fundamental Sciences, Islamic Republic Iran

Solution Let

$$\hat{f}(r) = \sum_{i=0}^{N-1} f(x) e^{\frac{-2r\pi i}{N}}$$

be the Fourier Transform of f for each $r \in Z_N$.

We have

$$\begin{aligned} \frac{1}{N^3} \sum_{r=0}^{N-1} \hat{f}(r) \hat{f}(r) \hat{f}(r)^{-1} &= \frac{1}{N^3} \sum_r \left(\sum_x f(x) e^{\frac{-2rx\pi i}{N}} \right) \left(\sum_y f(y) e^{\frac{-2ry\pi i}{N}} \right) \left(\sum_z f(z)^{-1} e^{\frac{2rz\pi i}{N}} \right) \\ &= \frac{1}{N^3} \sum_{x,y,z} f(x) f(y) f(z)^{-1} \left(\sum_r e^{\frac{-2r(x+y-z)\pi i}{N}} \right) \end{aligned}$$

We know that $\sum_r e^{\frac{-2r(x+y-z)\pi i}{N}} = N$ when $x+y=z$ and is 0 otherwise, so the latter is equal to

$$\frac{1}{N^2} \sum_{x,y} f(x) f(y) f(x+y)^{-1}.$$

Since $|f(x)f(y)f(x+y)^{-1} - 1| < \varepsilon$ for each x, y so we have:

$$\left| 1 - \frac{1}{N^3} \sum_r \hat{f}(r) \hat{f}(r) \hat{f}(r)^{-1} \right| < \varepsilon.$$

So by triangle inequality, we have:

$$\left| \frac{1}{N^3} \sum_r \hat{f}(r) \hat{f}(r) \hat{f}(r)^{-1} \right| > 1 - \varepsilon.$$

Also from Parseval identity we have

$$\frac{1}{N^2} \sum_r |\hat{f}(r)|^2 = \frac{1}{N} \sum_x |f(x)|^2 = 1$$

So we have

$$\frac{1}{N} \max_r |\hat{f}(r)| = \frac{1}{N} \max_r |f(r)| \cdot \left| \frac{1}{N^2} \sum_r |\hat{f}(r)|^2 \right| > \left| \frac{1}{N^2} \sum_r \hat{f}(r) \hat{f}(r) \right| > 1 - \varepsilon.$$

Therefore there is some r that $\hat{f}(r) > N(1 - \varepsilon)$. We will prove that for all $x \in Z_N$,

$$\left| f(x) - e^{\frac{2rx\pi i}{N}} \right| < 2\varepsilon.$$

We have

$$\begin{aligned}
\frac{1}{N} \left| f(x) - e^{\frac{2rx\pi i}{N}} \right| \hat{f}(r) &= \frac{1}{N} \left| f(x) - e^{\frac{2rx\pi i}{N}} \right| \left(\sum_y f(y) e^{\frac{-2ry\pi i}{N}} \right) \\
&= \left| \frac{1}{N} \left(\sum_y f(x)f(y) e^{\frac{-2ry\pi i}{N}} - \sum_y f(x+y) e^{\frac{-2r(y-x)\pi i}{N}} \right) \right| \\
&< \left| \frac{1}{N} \left(\sum_y f(x+y) e^{\frac{-2ry\pi i}{N}} - \sum_y f(x+y) e^{\frac{-2r(y-x)\pi i}{N}} \right) \right| + \varepsilon = \varepsilon
\end{aligned}$$

We used fact that $|f(x+y) - f(x) - f(y)| < \varepsilon$ and $\sum_y f(x+y) e^{\frac{-2ry\pi i}{N}} = \sum_y f(x+y) e^{\frac{-2r(y-x)\pi i}{N}}$ with a substitution of $y \rightarrow x+y$.

Hence

$$\left| f(x) - e^{\frac{2rx\pi i}{N}} \right| < \varepsilon \frac{N}{\hat{f}(r)} < \frac{\varepsilon}{1-\varepsilon} < 2\varepsilon.$$

So $g(x) = e^{\frac{2rx\pi i}{N}}$ is the answer.

Problem 10(Team) Marking Scheme

N _o	Steps	Points
1.	For Fourier transform	3pts
2.	For $\frac{1}{N^3} \sum_{r=0}^{N-1} \hat{f}(r) \hat{f}(r) \hat{f}(r)^{-1} = \frac{1}{N^3} \sum_{x,y,z} f(x)f(y)f(z)^{-1} \left(\sum_r e^{\frac{-2r(x+y-z)\pi i}{N}} \right)$	1pts
3.	For getting $\frac{1}{N^2} \sum_{x,y} f(x)f(y)f(x+y)^{-1}$	1pts
4.	For $\left \frac{1}{N^3} \sum_r \hat{f}(r) \hat{f}(r) \hat{f}(r)^{-1} \right > 1 - \varepsilon$	1pts
5.	For using Parseval identity and getting $\frac{1}{N^2} \sum_r \hat{f}(r) ^2 = \frac{1}{N} \sum_x \hat{f}(r) ^2 = 1$	1pts
6.	For part $\left f(x) - e^{\frac{2rx\pi i}{N}} \right < 2\varepsilon$	1pts
7.	For estimating $\frac{1}{N} \left f(x) - e^{\frac{2rx\pi i}{N}} \right \hat{f}(r) < \varepsilon$	1pts
8.	Getting $\left f(x) - e^{\frac{2rx\pi i}{N}} \right < 2\varepsilon$ by using conditions and giving answer	1pts